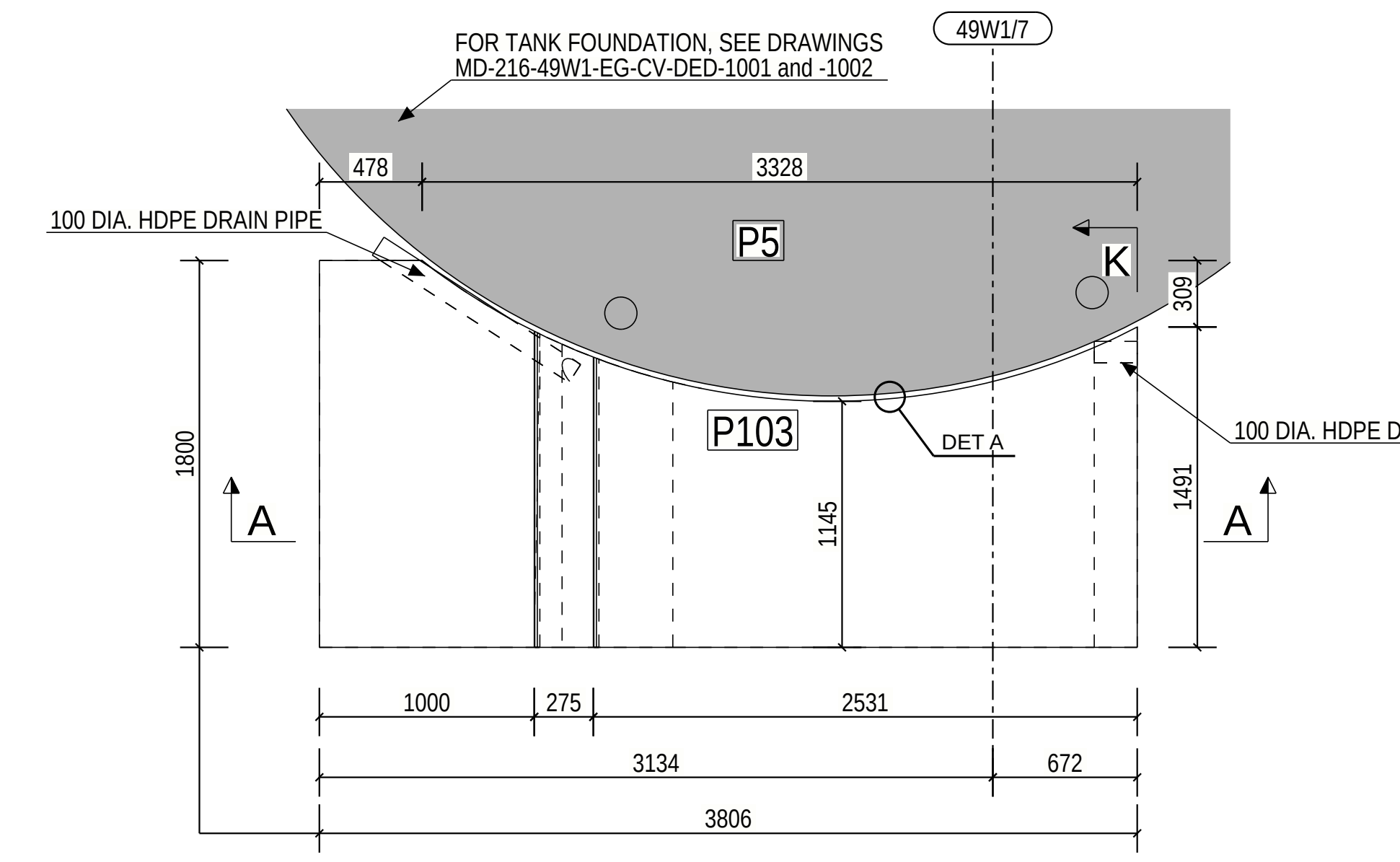
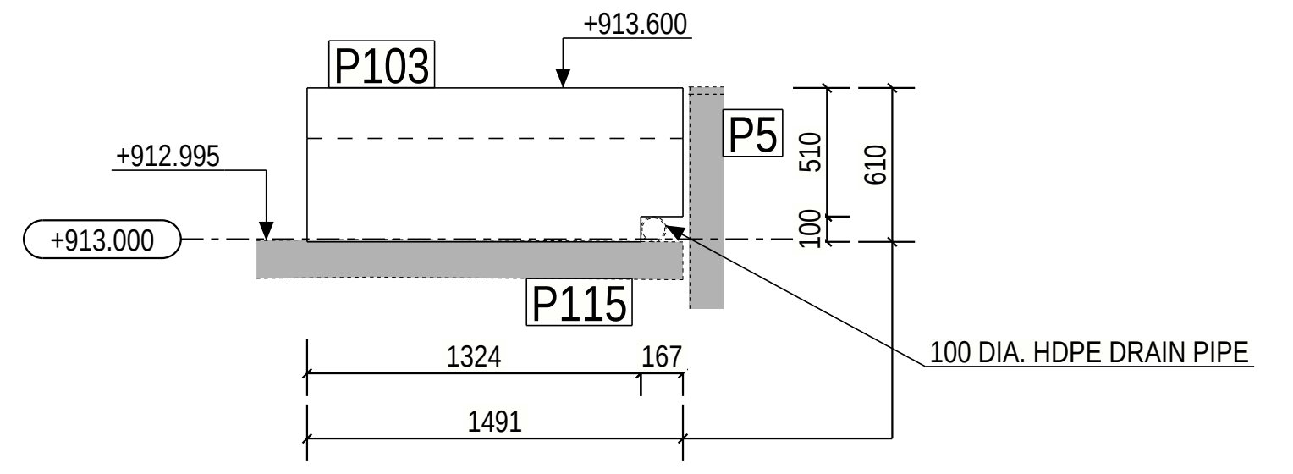
DRAWING NUMBER MD-216-49W1-EG-CV-DED-1013
REV D01

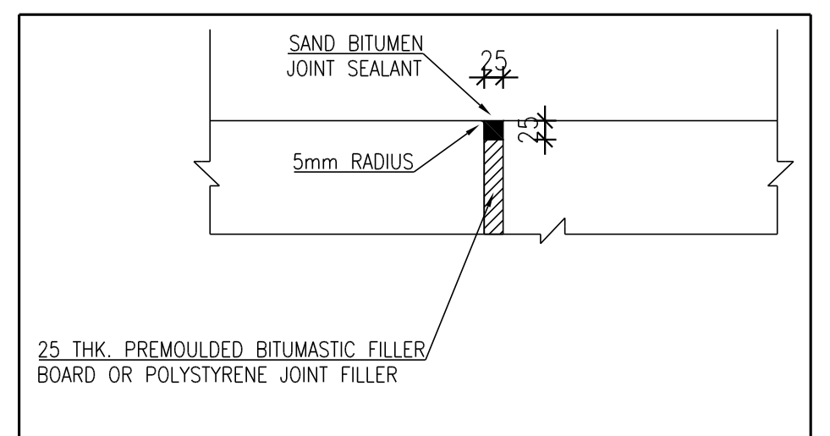


FOR SLAB DETAILS, SEE DRAWING MD-216-49W1-EG-CV-DED-1027

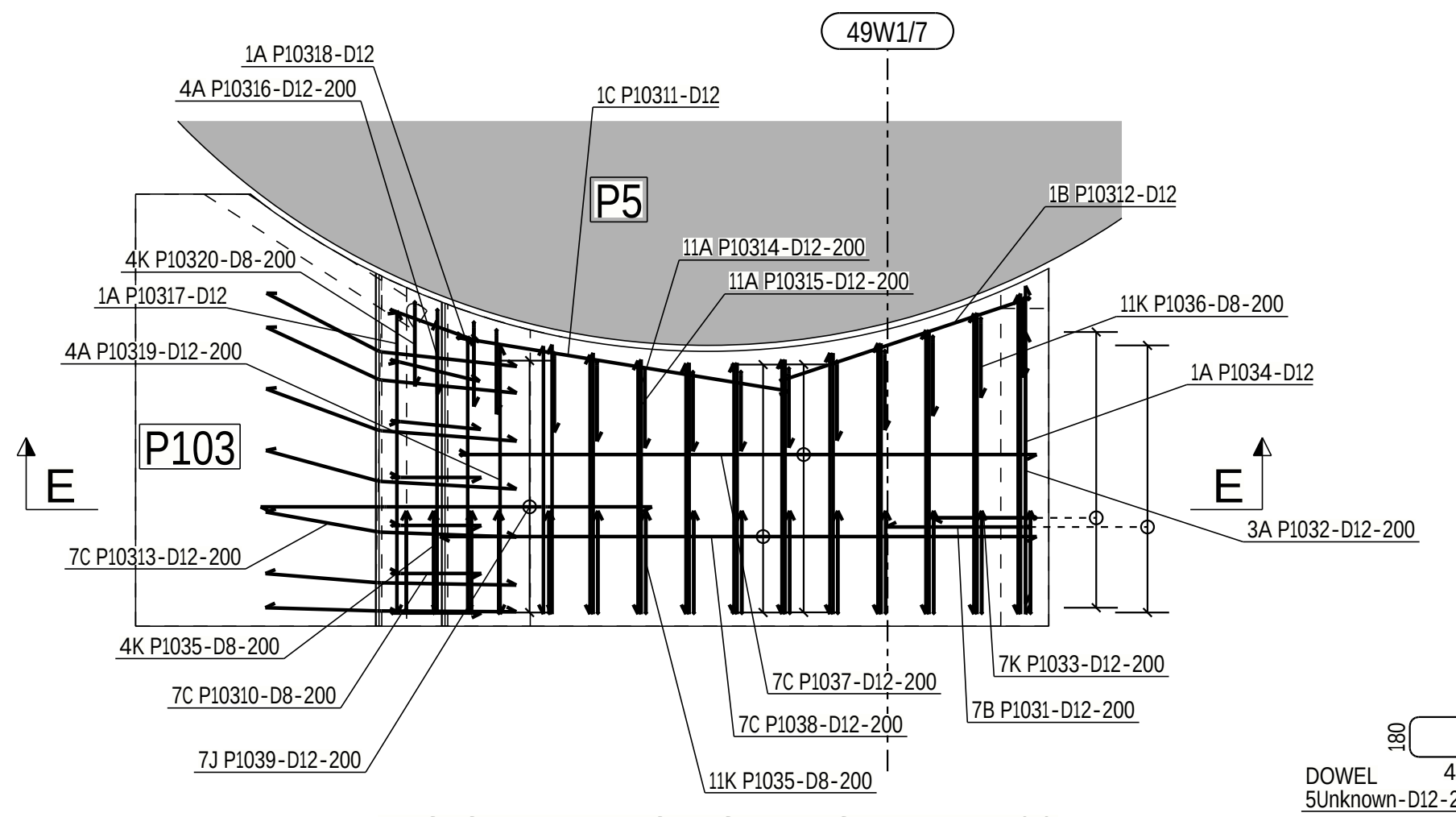
STAIRS MKD.P103
(SCALE 1:30)



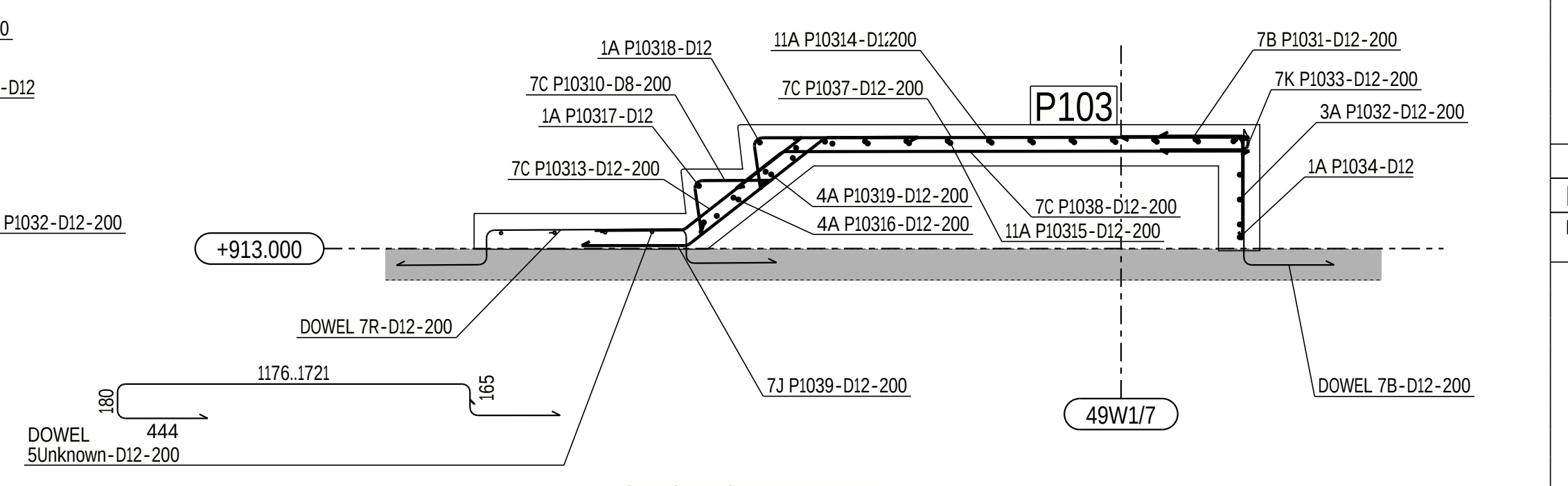
SECTION : K - K
(SCALE 1:30)


SAND, BITUMEN JOINT SEALANT
5mm RADIUS
25 THK. PREMOULDED BITUMASTIC FILLER BOARD OR POLYSTYRENE JOINT FILLER

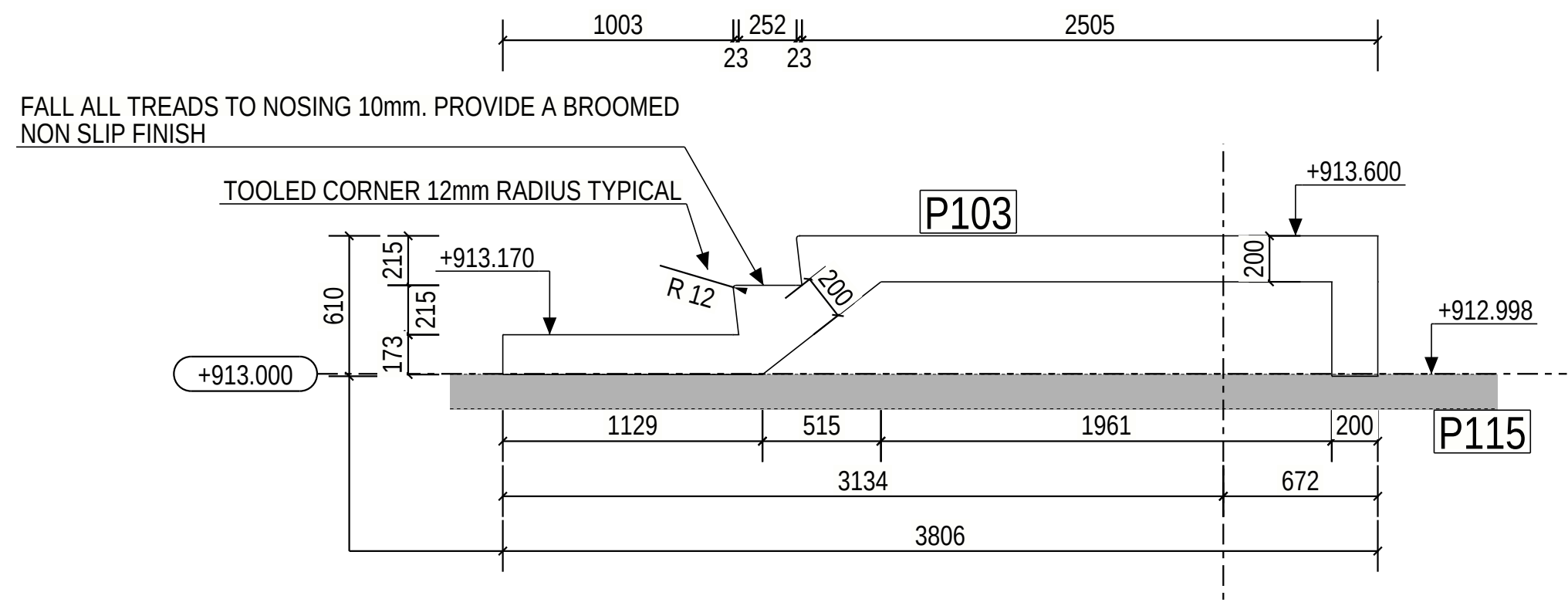
DETAIL OF ISOLATION GAP
SCALE 1:10
DET. A



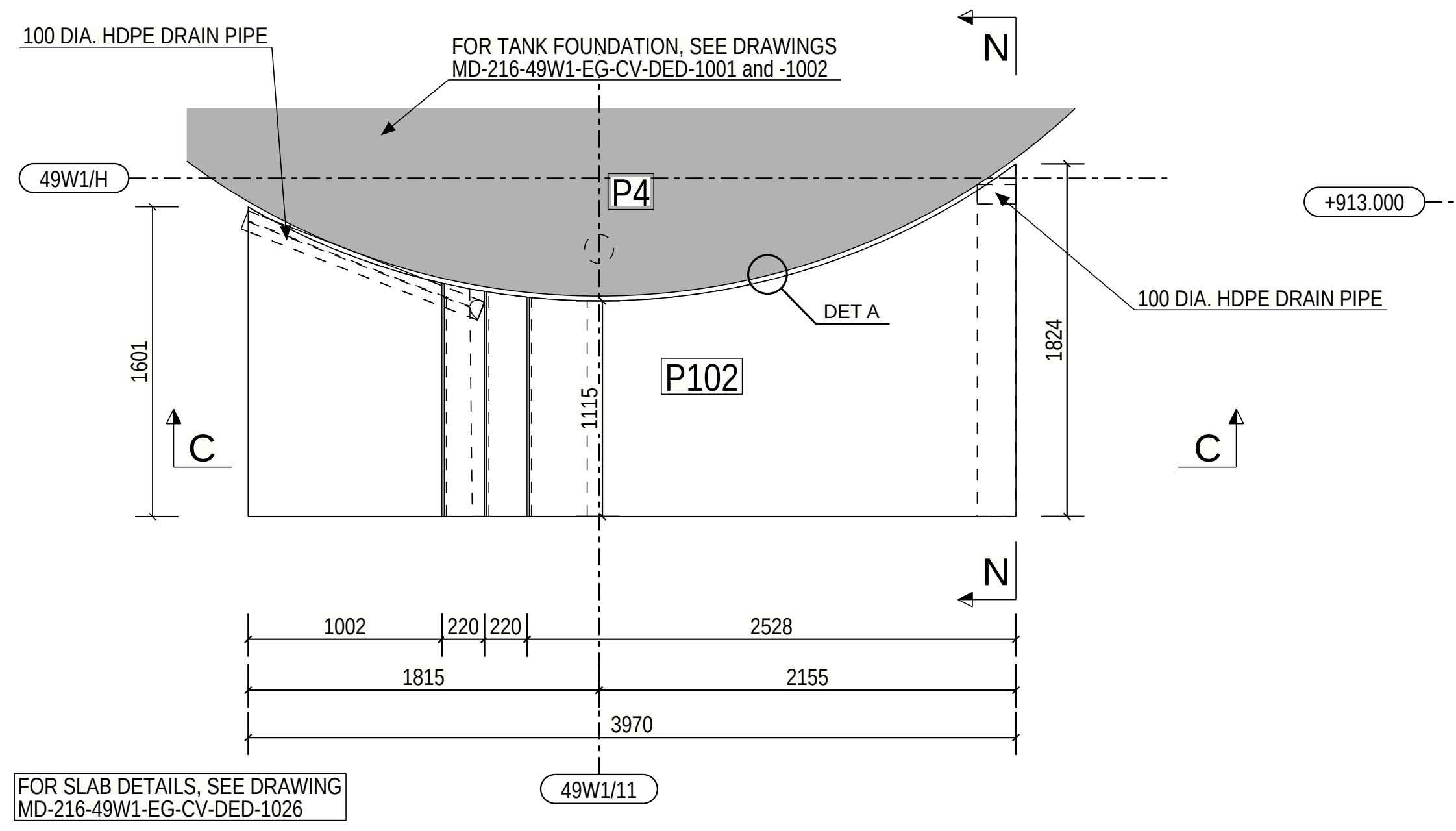
R.C.C. PLAN FOR STAIRS MKD.P103
(SCALE 1:30)



SECTION : E - E
(SCALE 1:30)

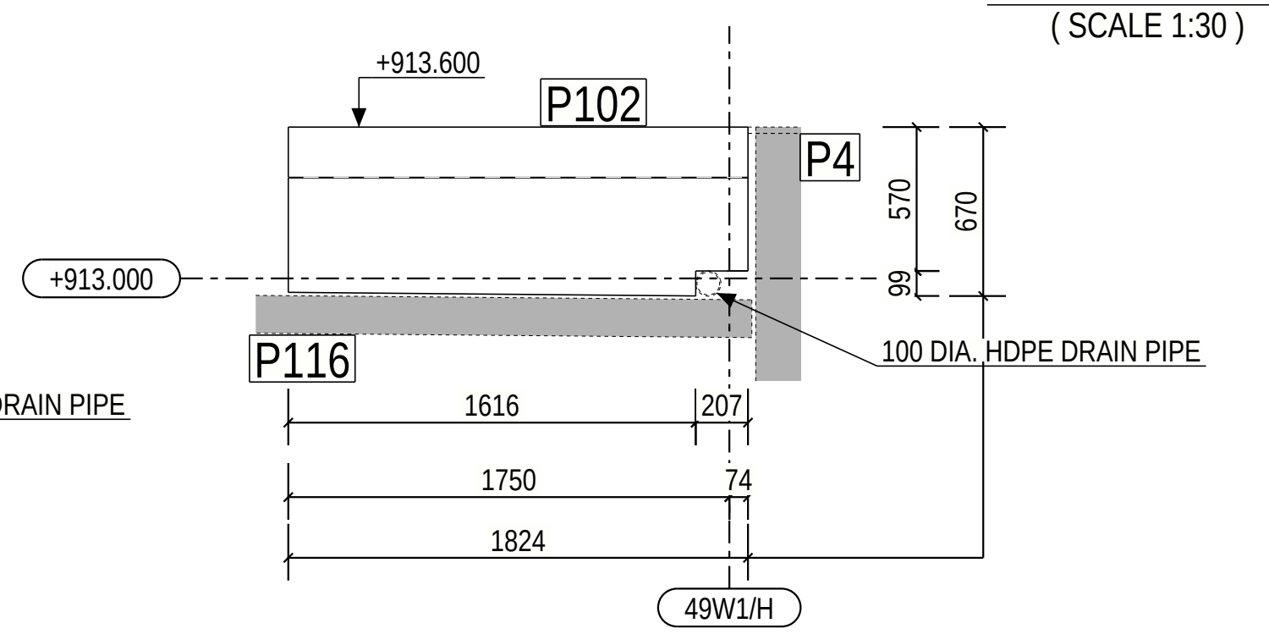

FALL ALL TREADS TO NOSING 10mm. PROVIDE A BROOMED NON SLIP FINISH
TOOLED CORNER 12mm RADIUS TYPICAL
R 12
P103
P115
+913.600
+913.170
+913.000
+912.998
1003
252
2505
1129
515
1961
200
3134
672
3806
49W1/7

SECTION : A - A
(SCALE 1:30)

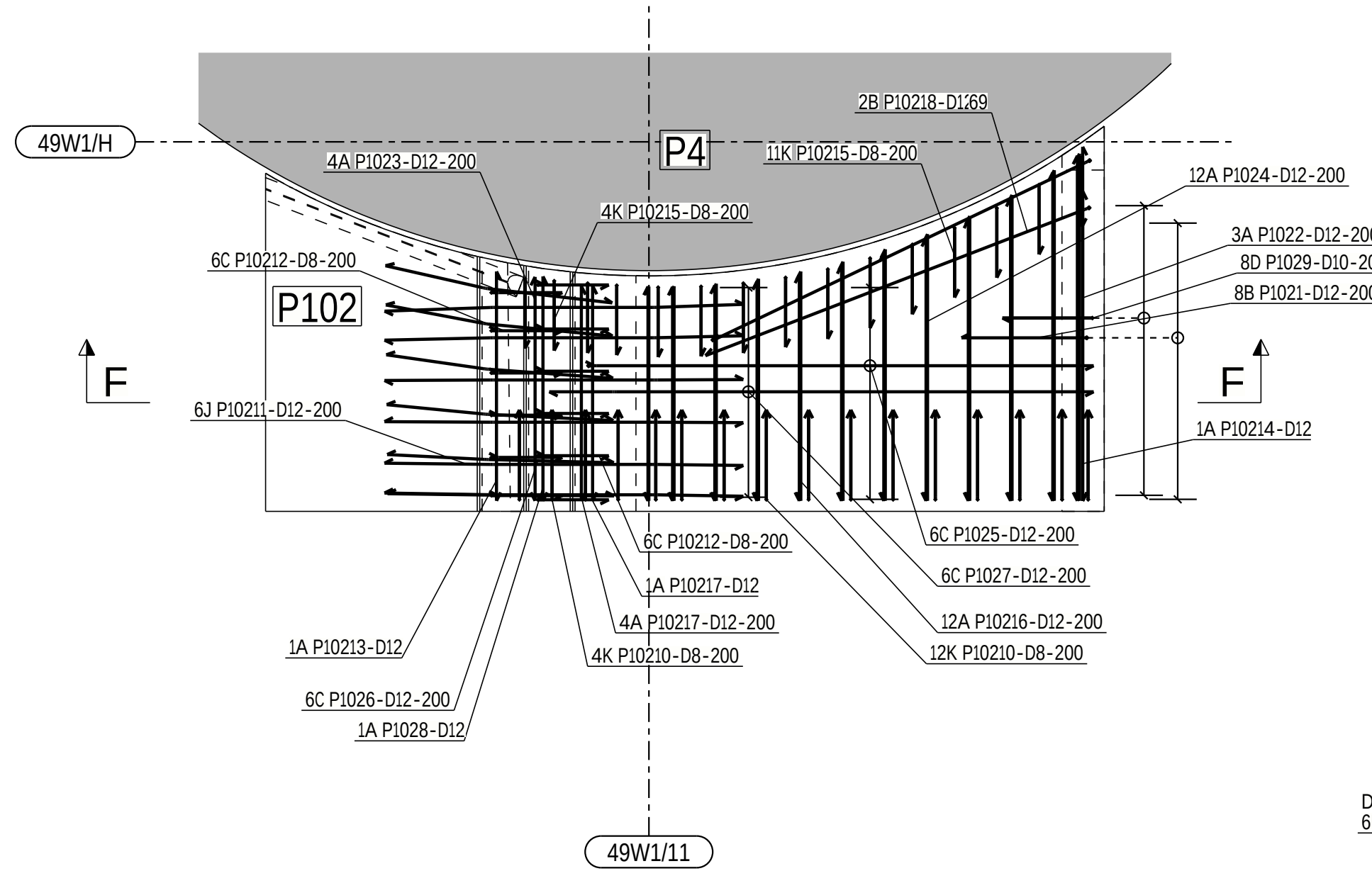


FOR SLAB DETAILS, SEE DRAWING MD-216-49W1-EG-CV-DED-1026

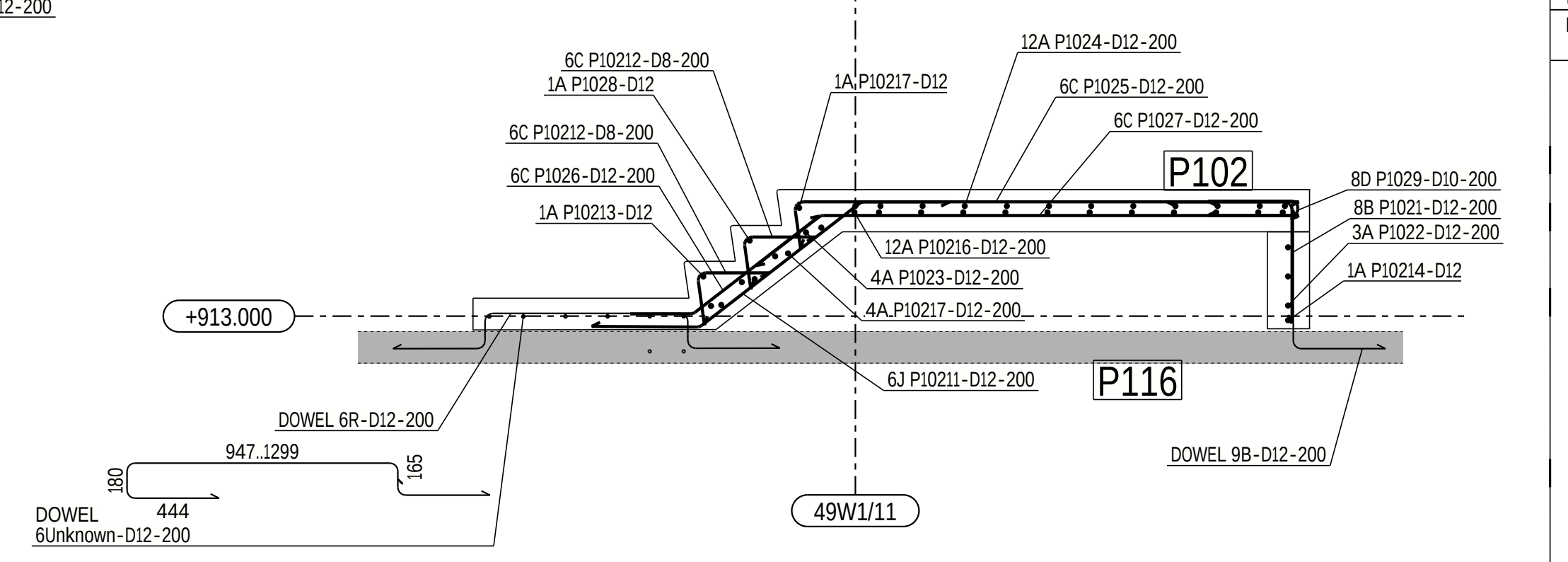
STAIRS MKD.P102
(SCALE 1:30)



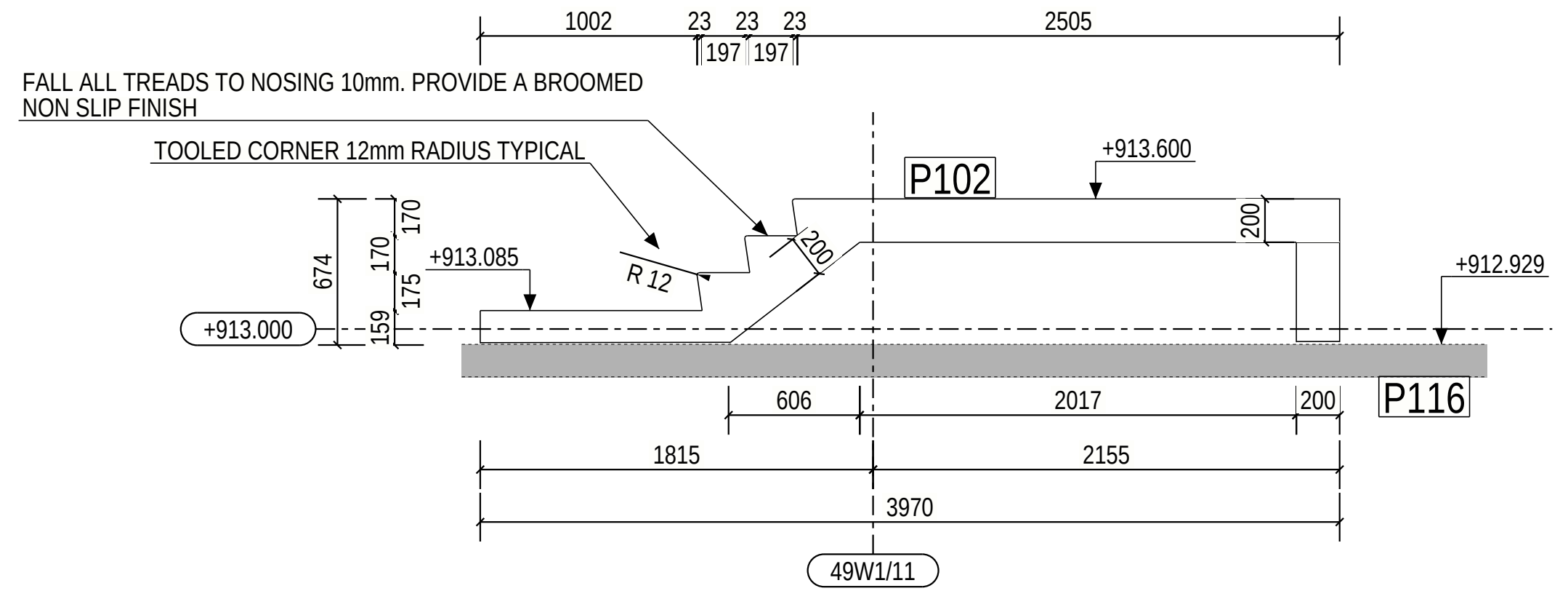
SECTION : N - N
(SCALE 1:30)



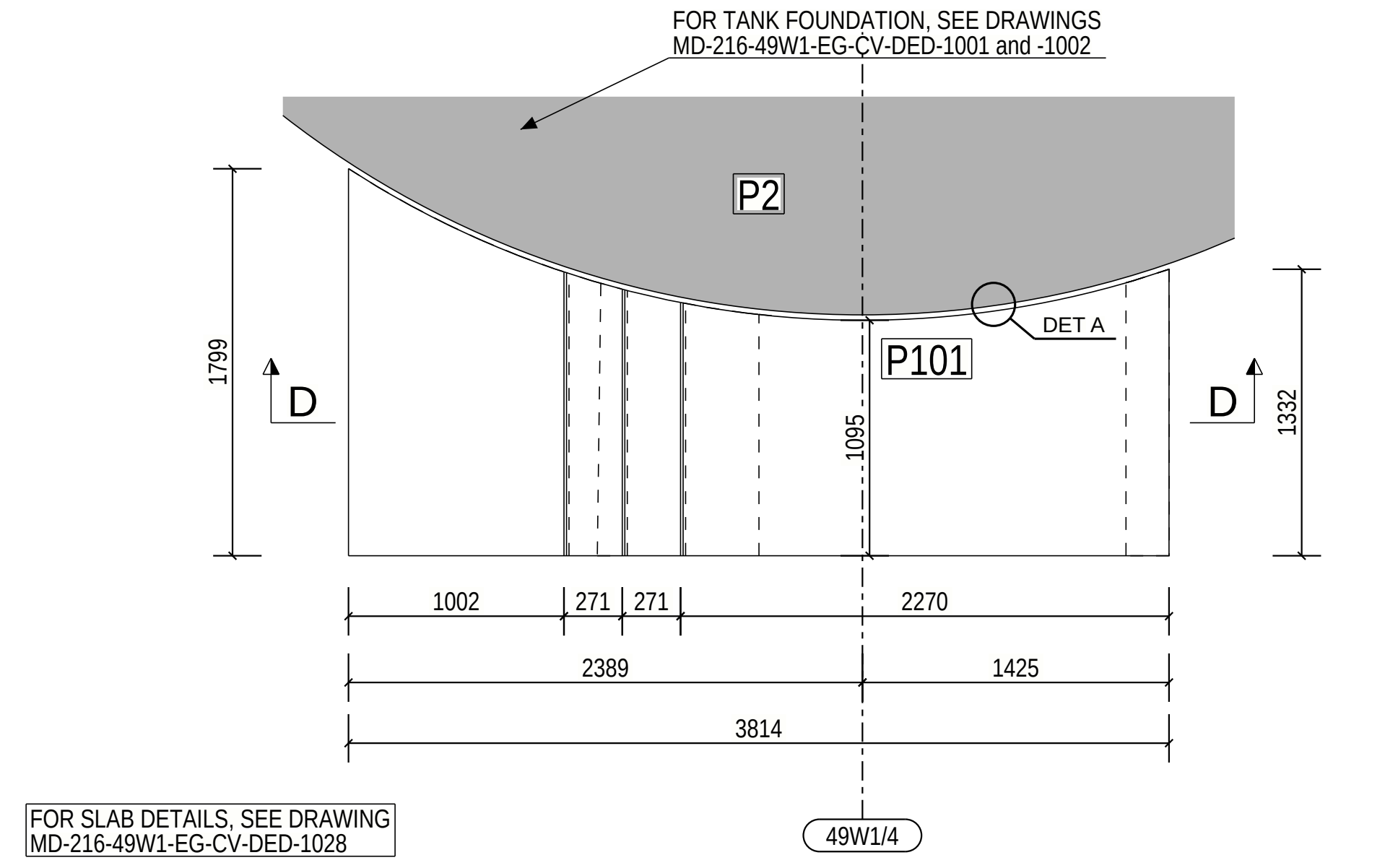
R.C.C. PLAN FOR STAIRS MKD.P102
(SCALE 1:30)



SECTION : F - F
(SCALE 1:30)

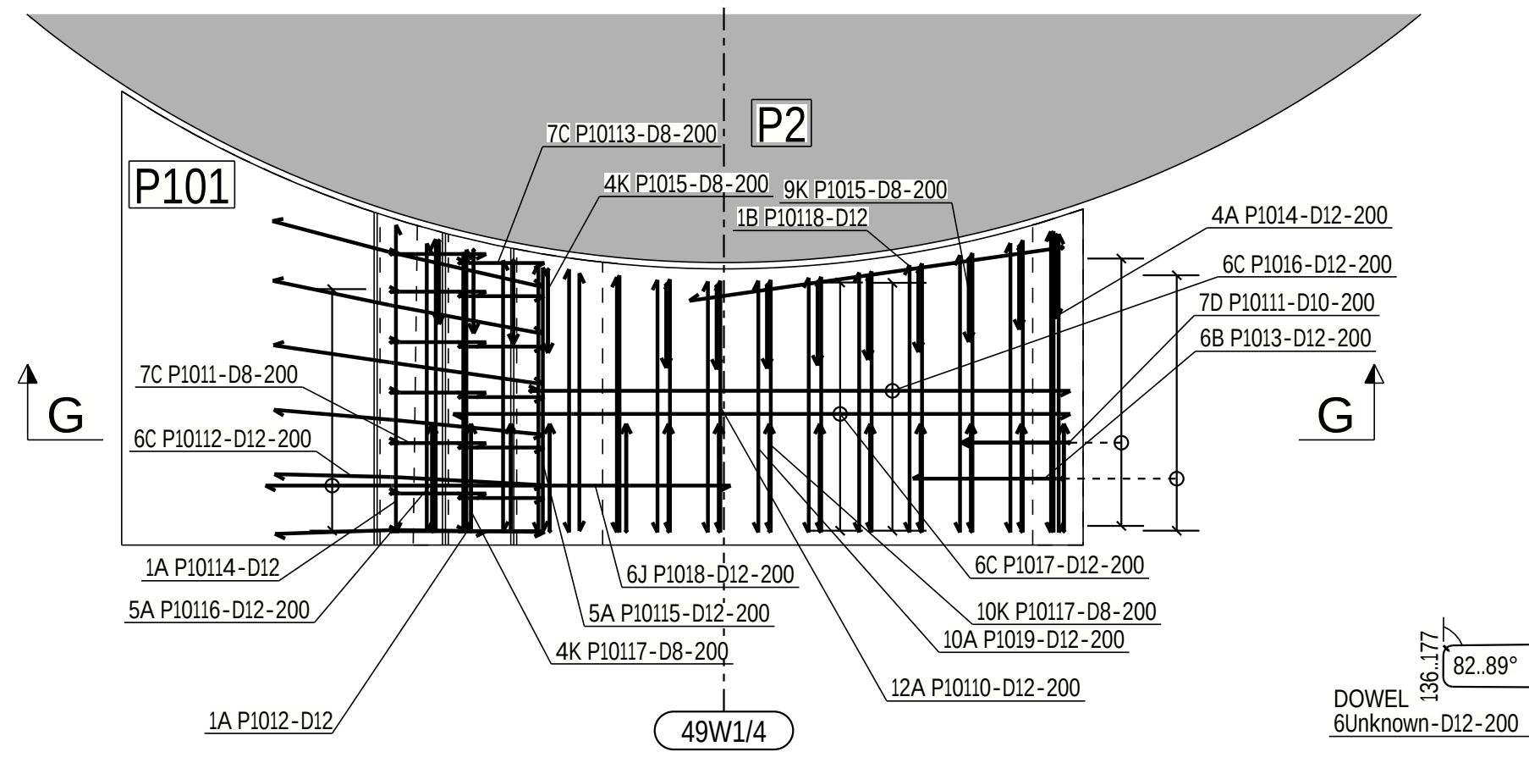

FALL ALL TREADS TO NOSING 10mm. PROVIDE A BROOMED NON SLIP FINISH
TOOLED CORNER 12mm RADIUS TYPICAL
R 12
P102
P116
+913.600
+913.085
+913.000
+912.929
1002
2505
606
2017
200
1815
2155
3970
49W1/11

SECTION : C - C
(SCALE 1:30)

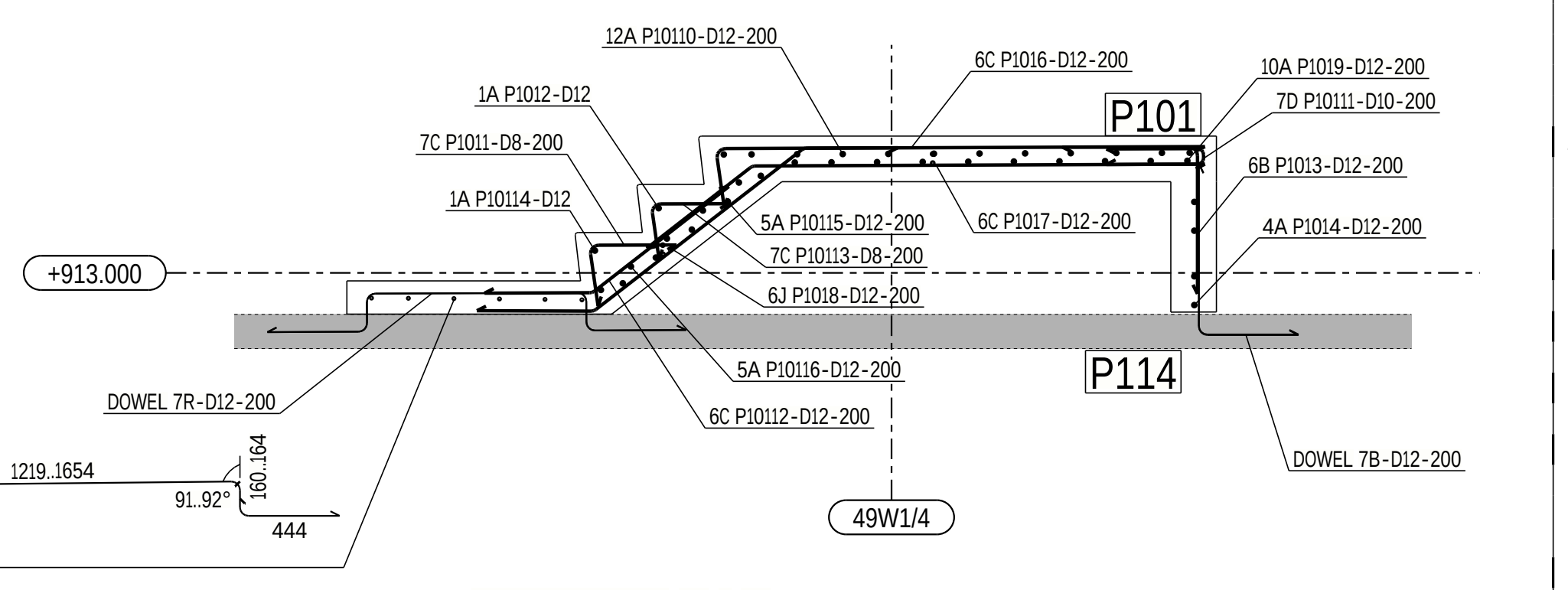

FOR TANK FOUNDATION, SEE DRAWINGS MD-216-49W1-EG-CV-DED-1001 and -1002
P2
P101
DET A
D
1799
1095
1332
1002
271
2270
2389
1425
3814
49W1/4

FOR SLAB DETAILS, SEE DRAWING MD-216-49W1-EG-CV-DED-1028

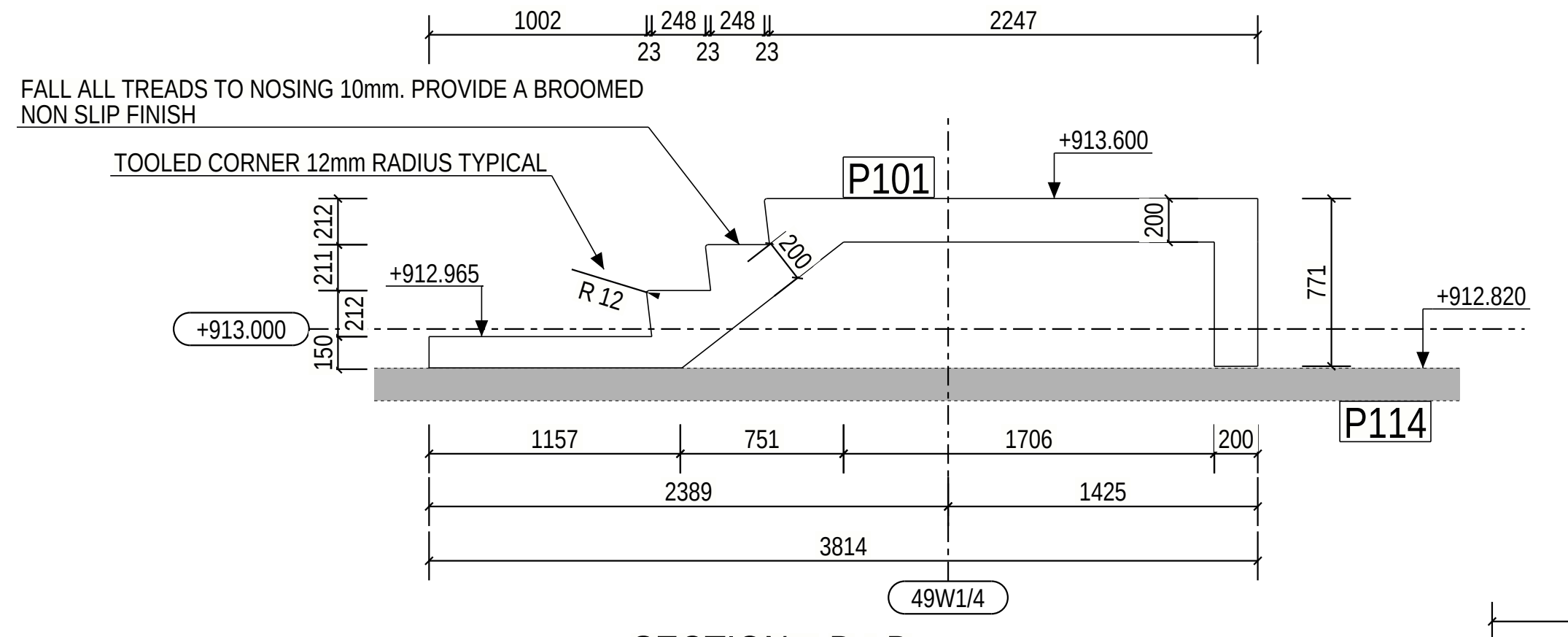
STAIRS MKD.P101
(SCALE 1:30)



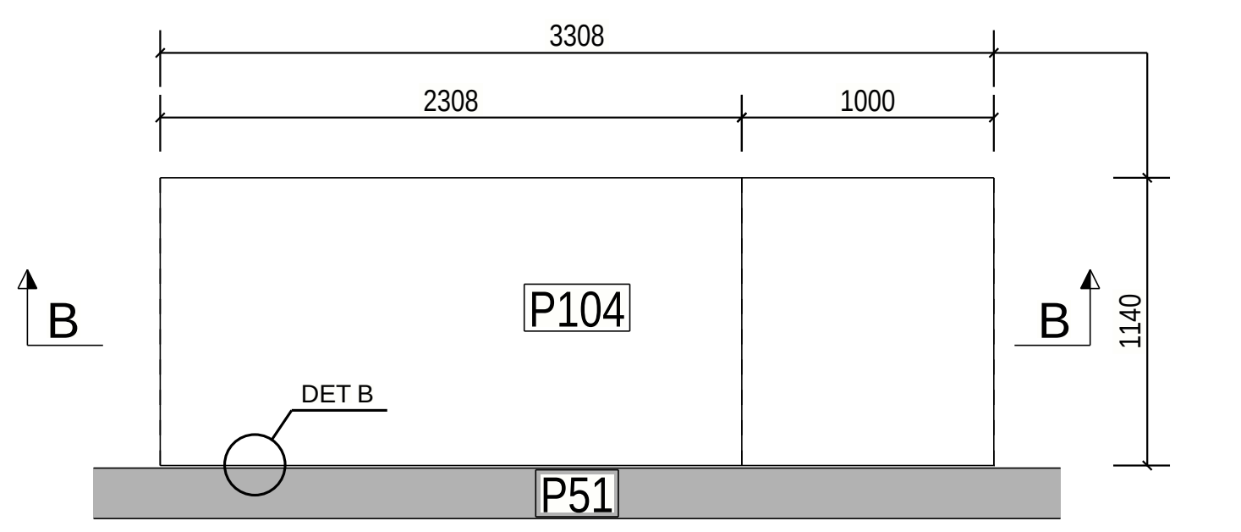
R.C.C. PLAN FOR STAIRS MKD.P101
(SCALE 1:30)



SECTION : G - G
(SCALE 1:30)

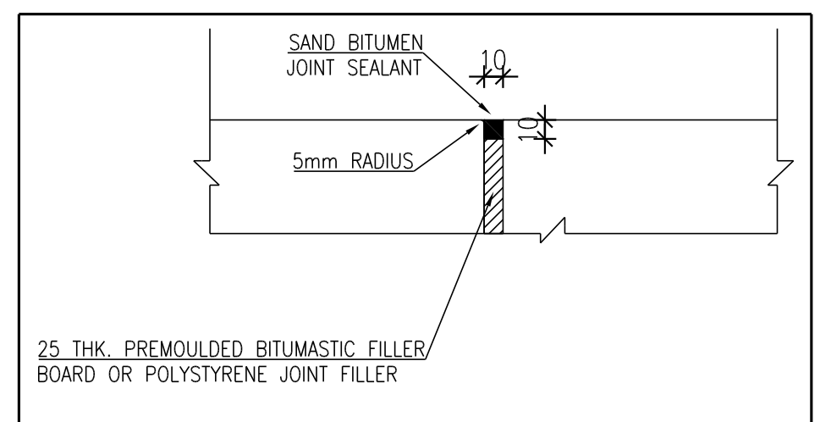

FALL ALL TREADS TO NOSING 10mm. PROVIDE A BROOMED NON SLIP FINISH
TOOLED CORNER 12mm RADIUS TYPICAL
R 12
P101
P114
+913.600
+912.965
+913.000
+912.820
1002
248
2247
1157
751
1706
200
2389
1425
3814
49W1/4

SECTION : D - D
(SCALE 1:30)

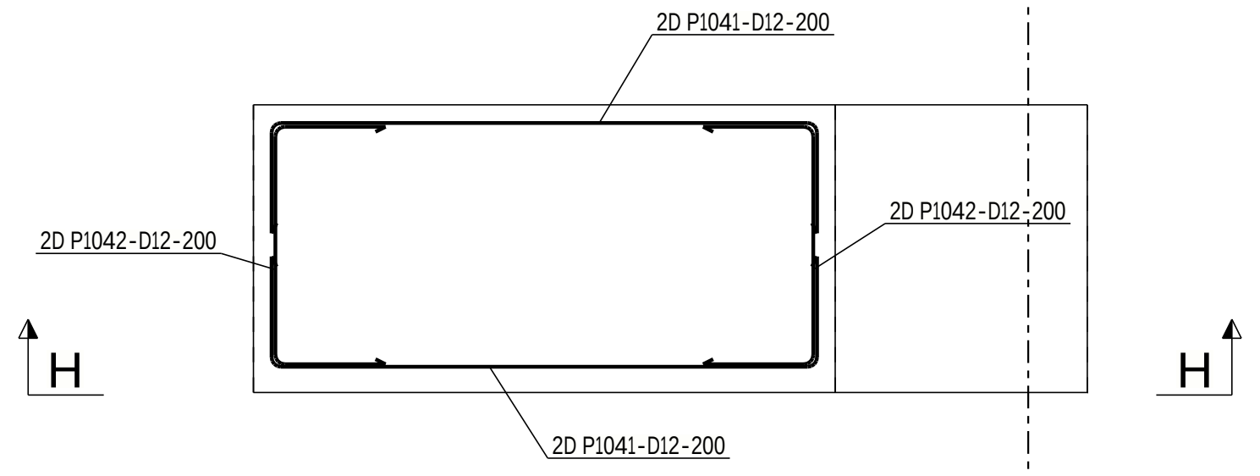


STAIRS MKD.P104
(SCALE 1:30)

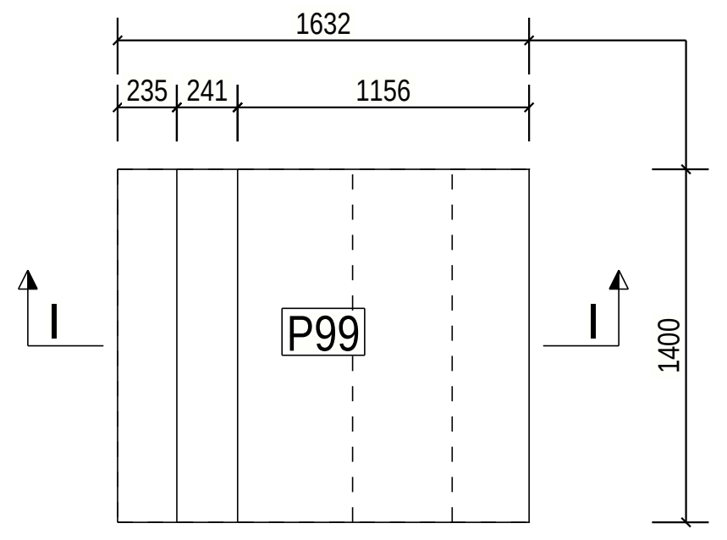
FOR SLAB DETAILS, SEE DRAWING MD-216-49W1-EG-CV-DED-1028


SAND, BITUMEN JOINT SEALANT
5mm RADIUS
25 THK. PREMOULDED BITUMASTIC FILLER BOARD OR POLYSTYRENE JOINT FILLER

DETAIL OF ISOLATION GAP
SCALE 1:10
DET. B

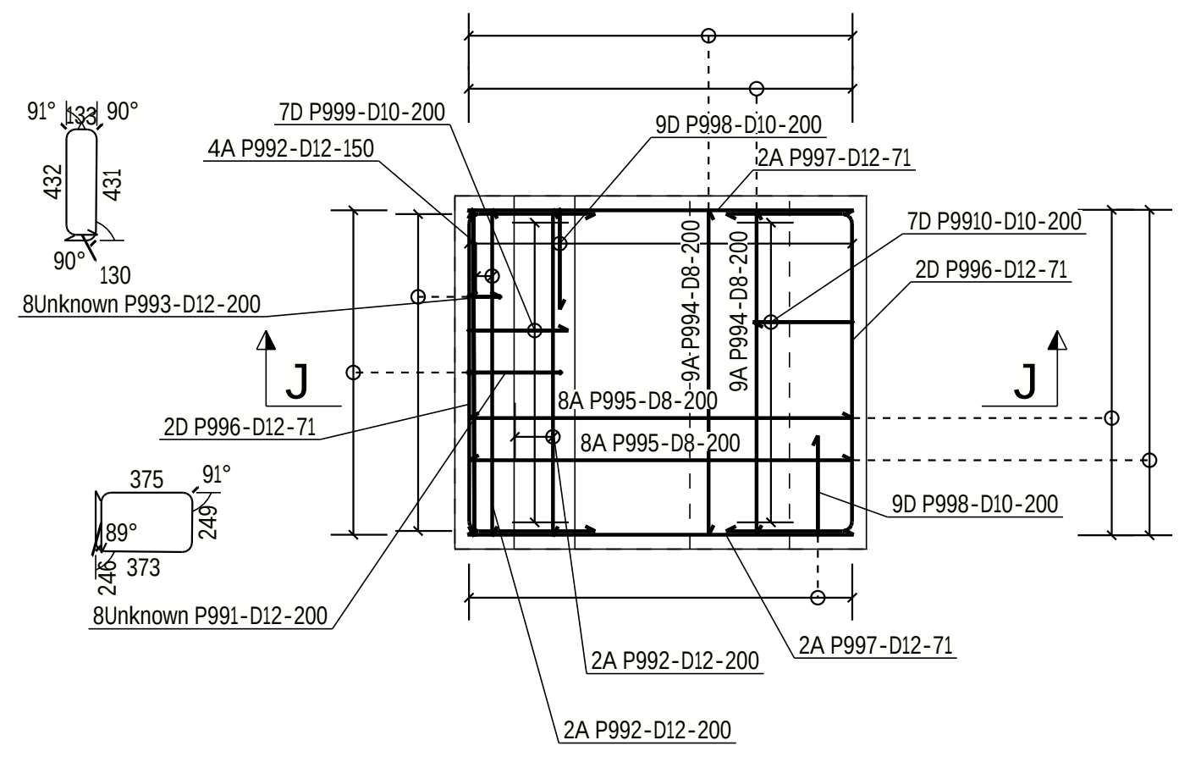


R.C.C. PLAN FOR STAIRS MKD.P104
(SCALE 1:30)



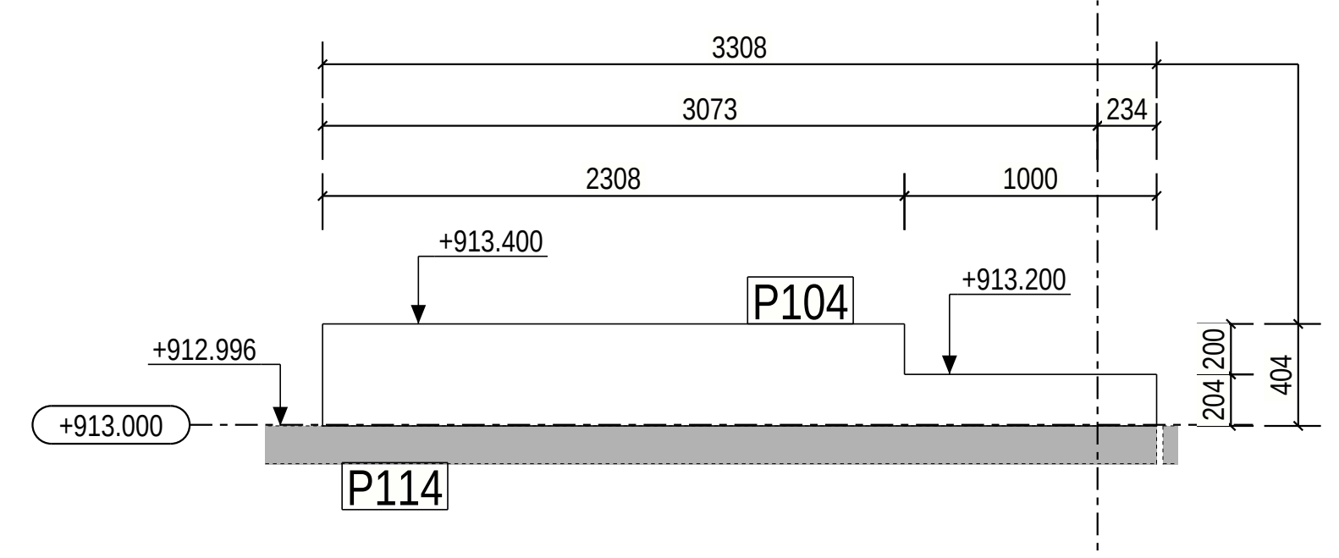
STAIRS MKD. P99
(SCALE 1:30)

FOR SLAB DETAILS, SEE DRAWING MD-216-49W1-EG-CV-DED-1029

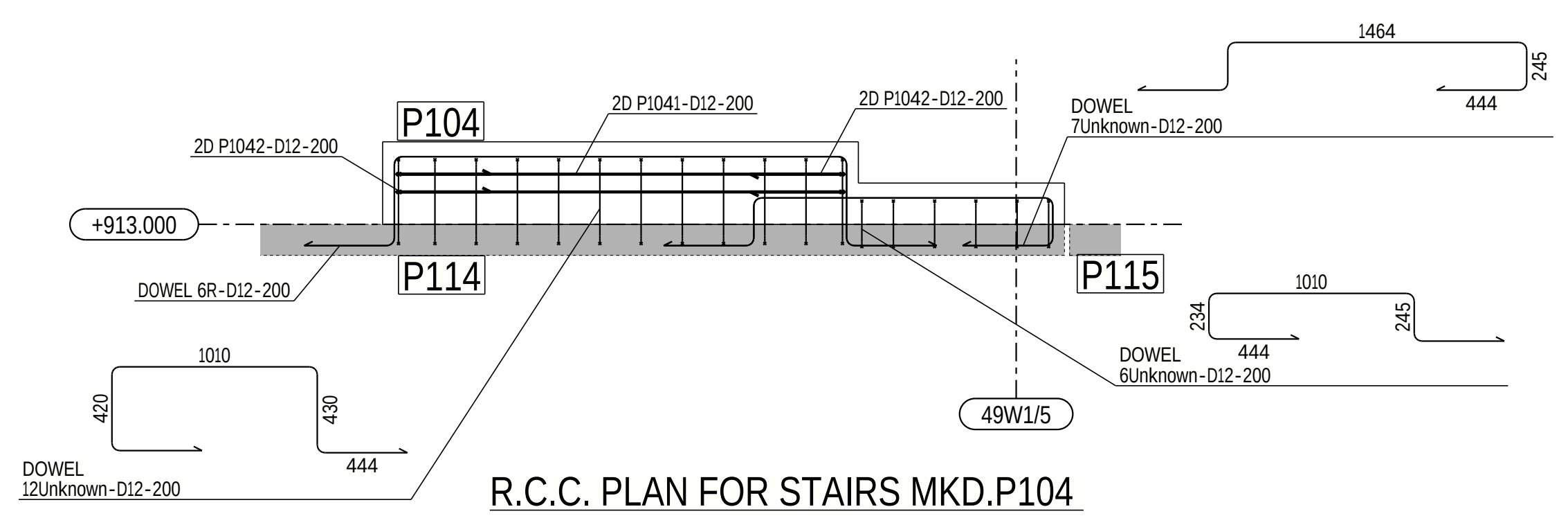


R.C.C. PLAN FOR STAIRS MKD.P99
(SCALE 1:30)

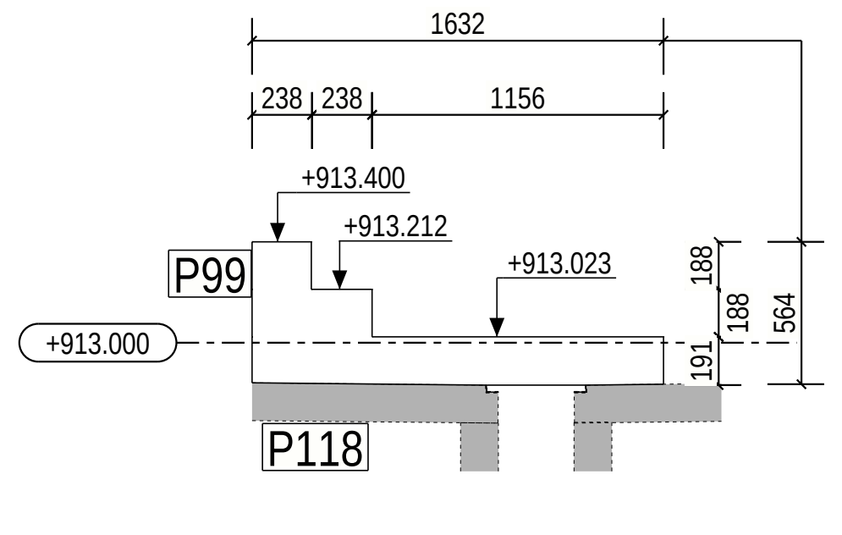
FOR LOCATION OF STAIRS, SEE DRAWING MD-216-49W1-EG-CV-GAD-1001, -1002 and -1003



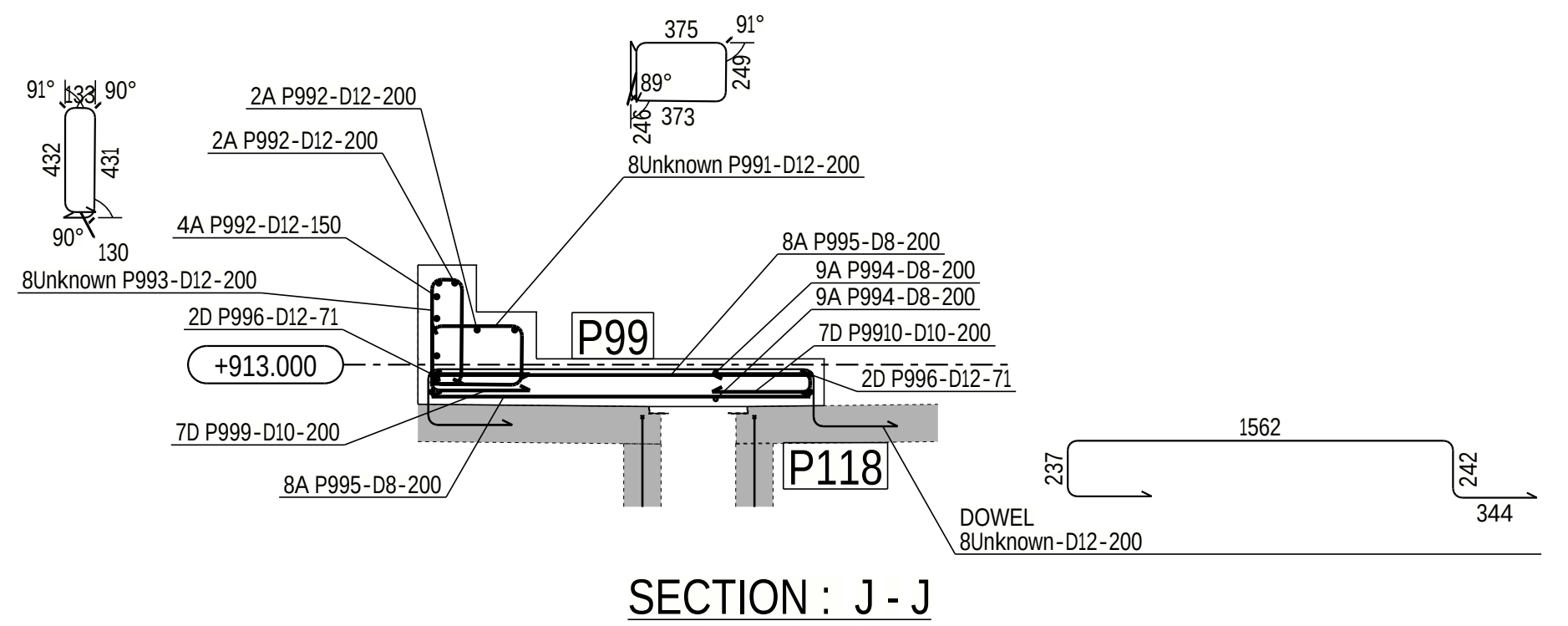
SECTION : B - B
(SCALE 1:30)



R.C.C. PLAN FOR STAIRS MKD.P104
(SCALE 1:30)



SECTION : I - I
(SCALE 1:30)



SECTION : J - J
(SCALE 1:30)

CONCRETE INFORMATION

Concretes used	C35
Concrete cover 1	75±10mm
Concrete cover 2	50±10mm

REINFORCING BAR LIST — CAST UNIT CODE: P99 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
**	P991	8	B500B	12	1370		9.7										
A	P992	8	B500B	12	1300		9.2	1300									
**	P993	8	B500B	12	1140		8.1										
A	P994	18	Undefined	8	1290		9.2	1290									
A	P995	16	Undefined	8	1520		9.6	1522									
D	P996	4	B500B	12	2220		7.9	507	1272	507						60	
A	P997	4	B500B	12	1530		5.4	1532									
D	P998	18	B500B	10	870		9.7	400	122	400						40	
D	P999	7	B500B	10	830		3.6	400	77	400						40	
D	P9910	7	B500B	10	830		3.6	400	83	400						40	

REINFORCING BAR TOTAL WEIGHT [kg] 76.2

REINFORCING BAR LIST — CAST UNIT CODE: P101 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
C	P1011	7	B500B	8	630		1.7	382	267				98			60	
A	P1012	1	B500B	12	1120		1.0	1121									
B	P1013	6	B500B	12	1220		6.5	651	600				90			60	
A	P1014	4	B500B	12	1180		4.2	1182									
K	P1015	13	B500B	8	680		3.5	335	335							32	
C	P1016	6	B500B	12	2380		12.7	2147	260				98			60	
C	P1017	6	B500B	12	2570		13.7	590	1987				38			60	
J	P1018	6	B500B	12	2080		11.1	515	1171	413			38			60	
A	P1019	10	B500B	12	1190		9.4	1193									
15	AT GROUP	10	PCS	...	990	22	...	995									
A	P10110	12	B500B	12	1190		11.3	1194									
1	AT GROUP	12	PCS	...	990	18	...	997									
D	P10111	7	B500B	10	890		3.9	432	80	432						40	
C	P10112	6	B500B	12	1260		6.6	493	773				38			60	
2	AT GROUP	6	PCS	...	1230	6	...	475	760				37				
C	P10113	7	B500B	8	560		1.6	342	242				98			60	
A	P10114	1	B500B	12	1210		1.1	1219									
A	P10115	5	B500B	12	1140		4.8	1144									
2	AT GROUP	5	PCS	...	1020	30	...	1022									
A	P10116	5	B500B	12	1190		5.0	1198									
1	AT GROUP	5	PCS	...	1080	28	...	1081									
K	P10117	14	B500B	8	860		4.8	430	430							40	
B	P10118	1	B500B	12	2120		1.9	651	1500				90			60	

REINFORCING BAR TOTAL WEIGHT [kg] 104.8

REINFORCING BAR LIST — CAST UNIT CODE: P102 — PCS: 2

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
B	P1021	8	B500B	12	1150		8.2	584	600				90			60	
A	P1022	3	B500B	12	1670		4.5	1677									
A	P1023	4	B500B	12	1080		3.7	1086									
7	AT GROUP	4	PCS	...	1000	27	...	1004									
A	P1024	12	B500B	12	1640		13.1	1643									
1	AT GROUP	12	PCS	...	1010	57	...	1014									
C	P1025	6	B500B	12	2590		13.8	2397	220				98			60	
C	P1026	6	B500B	12	1240		6.6	487	763				38			60	
2	AT GROUP	6	PCS	...	1220	4	...	475	760				38				
C	P1027	6	B500B	12	2650		14.2	390	2272				38			60	
A	P1028	1	B500B	12	1060		0.9	1062									
D	P1029	8	B500B	10	890		4.4	432	80	432						40	
K	P10210	16	B500B	8	860		5.5	430	430							40	
J	P10211	6	B500B	12	1900		10.1	515	975	421			38			60	
6	AT GROUP	6	PCS	...	1890	2	...	515	975	420			38				
C	P10212	12	B500B	8	570		2.7	341	247				98			60	
A	P10213	1	B500B	12	1080		1.0	1086									
A	P10214	1	B500B	12	1460		1.3	1466									
K	P10215	15	B500B	8	680		4.1	335	335							32	
A	P10216	12	B500B	12	1630		13.0	1634									
3	AT GROUP	12	PCS	...	1010	56	...	1012									
A	P10217	5	B500B	12	1010		4.5	1018									
B	P10218	2	B500B	12	2430		4.3	471	1990				90			60	
3	AT GROUP	2	PCS	...	2390	40	...	471	1953				90				

REINFORCING BAR TOTAL WEIGHT [kg] 115.9

REINFORCING BAR LIST — CAST UNIT CODE: P103 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
B	P1031	7	B500B	12	1070		6.7	507	600				90			60	
A	P1032	3	B500B	12	1360		3.6	1368									
K	P1033	7	B500B	12	890		5.6	434	434							40	
A	P1034	1	B500B	12	1130		1.0	1130									
K	P1035	15	B500B	8	860		5.1	430	430							40	
K	P1036	11	B500B	8	680		3.0	335	335							32	
C	P1037	7	B500B	12	2630		16.4	2405	250				98			60	
C	P1038	7	B500B	12	2540		15.8	290	2262				38			60	
J	P1039	7	B500B	12	1810		11.3	515	853	452			38			60	
C	P10310	7	B500B	8	620		1.7	393	247				98			60	
10	AT GROUP	7	PCS	...	600	3	...	376	247				98				
C	P10311	1	B500B	12	1630		1.5	1404	250				98			60	
B	P10312	1	B500B	12	1460		1.3	398	1100				90			60	
C	P10313	7	B500B	12	1250		7.6	534	730				42			60	
1	AT GROUP	7	PCS	...	1200	8	...	475	729				38				
A	P10314	11	B500B	12	1330		11.0	1336									
1	AT GROUP	11	PCS	...	1040	29	...	1045									
A	P10315	11	B500B	12	1330		11.0	1330									
1	AT GROUP	11	PCS	...	1040	29	...	1047									
A	P10316	4	B500B	12	1220		4.2	1223									
1	AT GROUP	4	PCS	...	1160	20	...	1168									
A	P10317	1	B500B	12	1260		1.1	1264									
A	P10318	1	B500B	12	1150		1.0	1158									
A	P10319	4	B500B	12	1160		4.0	1163									
1	AT GROUP	4	PCS	...	1060	33	...	1060									
K	P10320	4	B500B	8	720		1.1	355	355							32	

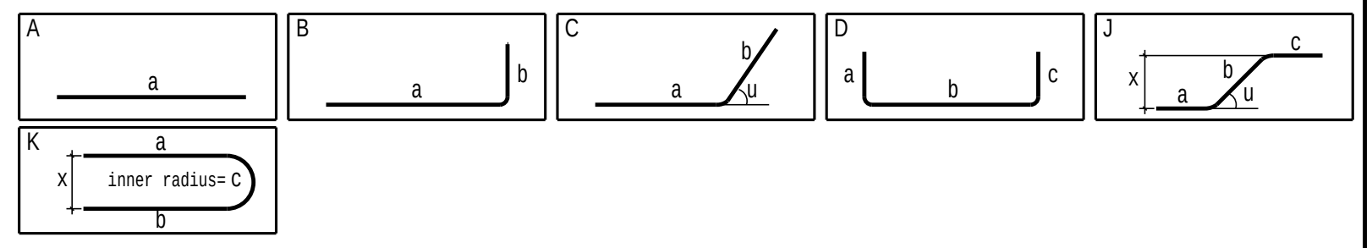
REINFORCING BAR TOTAL WEIGHT [kg] 114.1

REINFORCING BAR LIST — CAST UNIT CODE: P104 — PCS: 1

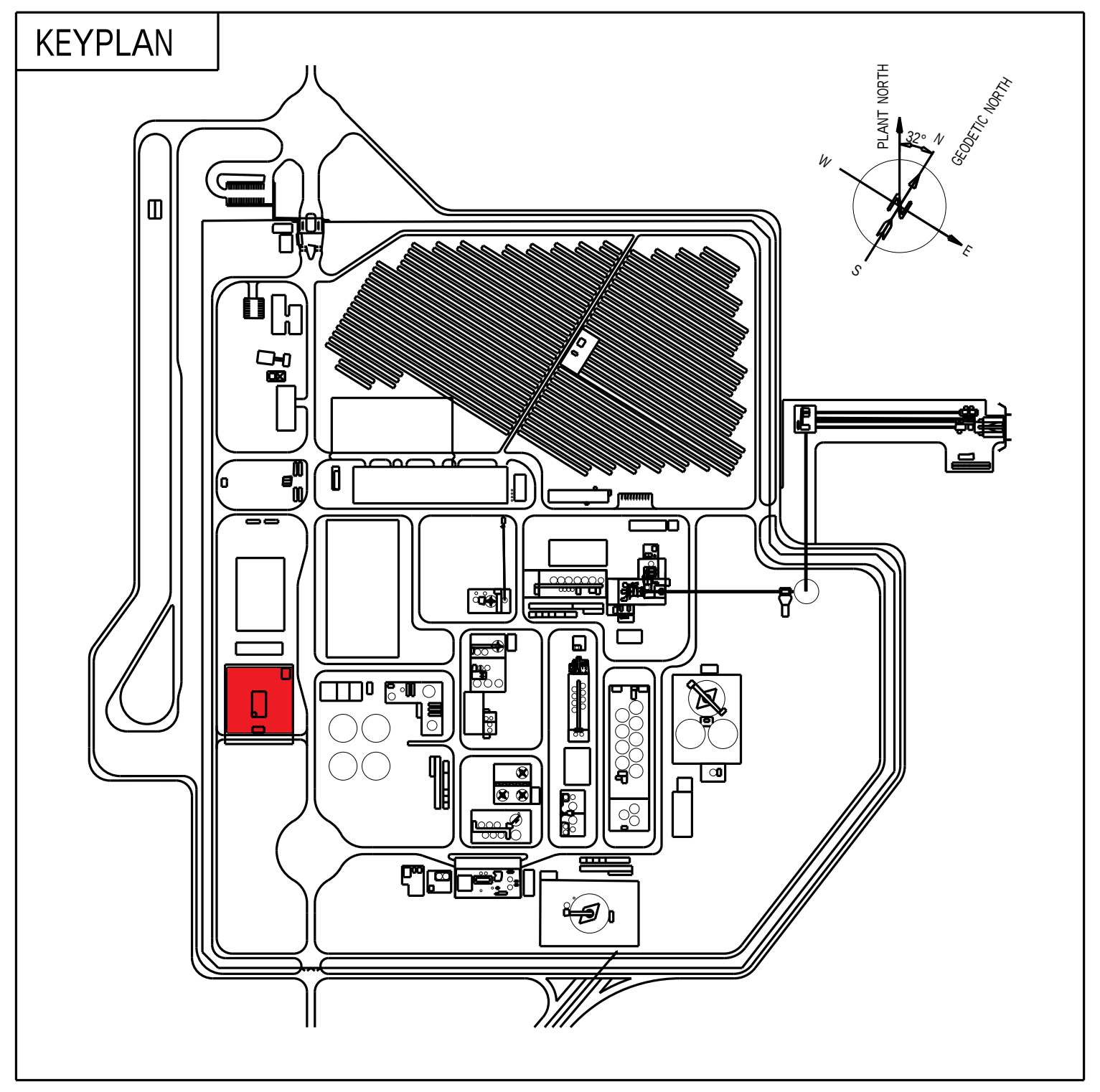
TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
D	P1041	4	B500B	12	3000		10.7	444	2177	444						60	
D	P1042	4	B500B	12	1780		6.3	444	955	444						60	

REINFORCING BAR TOTAL WEIGHT [kg] 17.0

REINFORCEMENT BENDING SHAPES:



KEYPLAN



LEGEND:

R.L. - REDUCED LEVEL
T.O.C. - TOP OF CONCRETE
B.O.F. - BOTTOM OF FOUNDATION
F.G.L. - FINISHED GROUND LEVEL
P.C.C. - PLAIN CEMENT CONCRETE
TYP - TYPICAL
MKD - MARKED
CL - CENTER LINE
BOTTOM REINF. (B)
TOP REINF. (T)
SBC - SAFE BEARING CAPACITY
MIN. - MINIMUM
REF. - REFERED
GA - GENERAL ARRANGEMENT
DWG. - DRAWING
R.C.C. - REINFORCED CEMENT CONCRETE

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS ARE IN METRES UNLESS OTHERWISE STATED
2. GRADE OF CONCRETE SHALL BE C35 (35MPa) UNLESS NOTED OTHERWISE & C15 (15MPa) FOR BLINDING CONCRETE.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615-GRADE 60 (fy=420MPa)
4. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS FOLLOWS:
a) FOUNDATION: 75 mm (BOTTOM & SIDE)
b) FOUNDATION: 50 mm (TOP)
c) COLUMN & PEDESTAL: 50 mm
5. MINIMUM LAP LENGTHS OF REINFORCEMENT SHALL BE 40XBAR DIAMETER
6. FOR STANDARD REINFORCEMENT DETAILS & GENERAL NOTES FOR CIVIL WORKS REFER TO DRAWING NUMBER MD-216-1000-EG-CV-DED-1005 & 1001
7. ANCHOR BOLTS AND EMBEDDED STEELWORK SHALL BE HOT DIP GALVANIZED
8. T.O.C. & TOP OF SCREED TOLERANCE SHALL BE WITHIN +/- 10MM OF DESIGN LEVEL
9. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION IS CONSIDERED AS 375 KN/SQ.M.
10. REBARS CAN BE MOVED LOCALLY BY +/- 25mm TO AVOID INTERFERENCE WITH EMBEDDED PARTS

DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-49W1-EG-CV-DED-1031	SLAB SECTOR GL D-G/1-4 DETAIL- AND REINFORCEMENT DRAWING		
MD-216-49W1-EG-CV-DED-1029	SLAB SECTOR GL D-G/8-12 DETAIL- AND REINFORCEMENT DRAWING	MD-216-49W1-EG-CV-DED-1030	SLAB SECTOR GL D-G/4-8 DETAIL- AND REINFORCEMENT DRAWING
MD-216-49W1-EG-CV-DED-1027	SLAB SECTOR GL G-J/5-10 DETAIL- AND REINFORCEMENT DRAWING	MD-216-49W1-EG-CV-DED-1028	SLAB SECTOR GL G-J/1-5 DETAIL- AND REINFORCEMENT DRAWING
MD-216-49W1-EG-CV-DED-1002	TANK FOUNDATIONS 49W1-TNK-003/007/009 DETAIL- AND REINFORCEMENT DRAWING	MD-216-49W1-EG-CV-DED-1026	SLAB SECTOR GL G-J/10-12 DETAIL- AND REINFORCEMENT DRAWING
MD-216-49W1-EG-CV-GAD-1003	1-10/A-E GENERAL ARRANGEMENT DRAWING	MD-216-49W1-EG-CV-DED-1001	TANK FOUNDATIONS 49W1-REA-015 & 49W1-REA-003 DETAIL- AND REINFORCEMENT DRAWING
MD-216-49W1-EG-CV-GAD-1001	1-7/E-J GENERAL ARRANGEMENT DRAWING	MD-216-49W1-EG-CV-GAD-1002	7-12/E-J GENERAL ARRANGEMENT DRAWING

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
D01	29.01.2024	AS-BUILT	JonRen	MauNot	LuiPiz	SouCha	
C01	01.03.2022	ISSUED FOR CONSTRUCTION	JonRen	MauNot	NanWu	LuiPiz	
A01	11.02.2022	ISSUED FOR APPROVAL	EvgPon	MauNot	NanWu	LuiPiz	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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OUTOTEC DRAWING NUMBER 900707_GAU01_AG11_CTB01_49013 — DOCUMENT ID OU100732645 — REVISION 2

DRAWN	EvgPon	11.02.2022
CHECKED	MauNot	11.02.2022
ENGINEER	NanWu	11.02.2022
APPROVED	LuiPiz	11.02.2022
EMPLOYER		
SCALE	1:30	A0

MANSOURAH - MASSARAH GOLD PROJECT
49W1 RAW WATER TREATMENT PLANT
STAIRS
DETAIL- AND REINFORCEMENT DRAWING

DRAWING NUMBER MD-216-49W1-EG-CV-DED-1013
REV D01

Tekla Structures